Multivariate Times Series and Analysis

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Voluntary Homework Paper

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Report

1. Summary the paper

Diego Känzig's 2021 paper described how people's expectations about future oil supply, not just sudden changes in current production, influence oil prices and the overall economy. The author used quick changes in oil futures prices around OPEC announcements as a way to measure "oil supply news shocks." His findings showed that bad news about future oil supply causes oil prices to go up right away and keep rising, while oil production slowly drops and storage levels go up. These shocks then lower industrial production worldwide and in the U.S., increase U.S. consumer prices, and weaken the value of the dollar, showing signs of stagflation. Looking at history, big events in the oil market, like the Iranian Revolution or the 2014 oil price drop, were partly because of changes in expectations about future supply. In summary, the expectations about future oil supply plays important role in how oil prices move and strongly affect the global economy.

2. Data and Variables

(1) Sample and frequency

We use the monthly replication dataset from Kaenzig (1974M01–2017M12), covering 528 observations and six endogenous variables.

(2) 6 Endogenous variables

Variables shown in Figure 1:

POIL (Real oil price): Pronounced volatility and cyclical swings; large run-ups and subsequent corrections are visible, consistent with well-known oil price episodes.

OILPROD (World oil production): Gradual upward drift with short-run fluctuations; occasional soft patches but a clear rising profile over the sample.

OILSTOCKS (World oil inventories): Strong secular increase early in the sample followed by slower accumulation.

WORLDIP (World industrial production): Persistent upward trend with short-term dips around global downturns; the growth path is relatively smooth.

IP (U.S. industrial production): Similar rising profile with cyclical setbacks; mid-sample slowdowns and late-sample rebounds are visible.

CPI (U.S. consumer price index): Steady upward trajectory consistent with a price level index.

(3) External instrument (proxy)

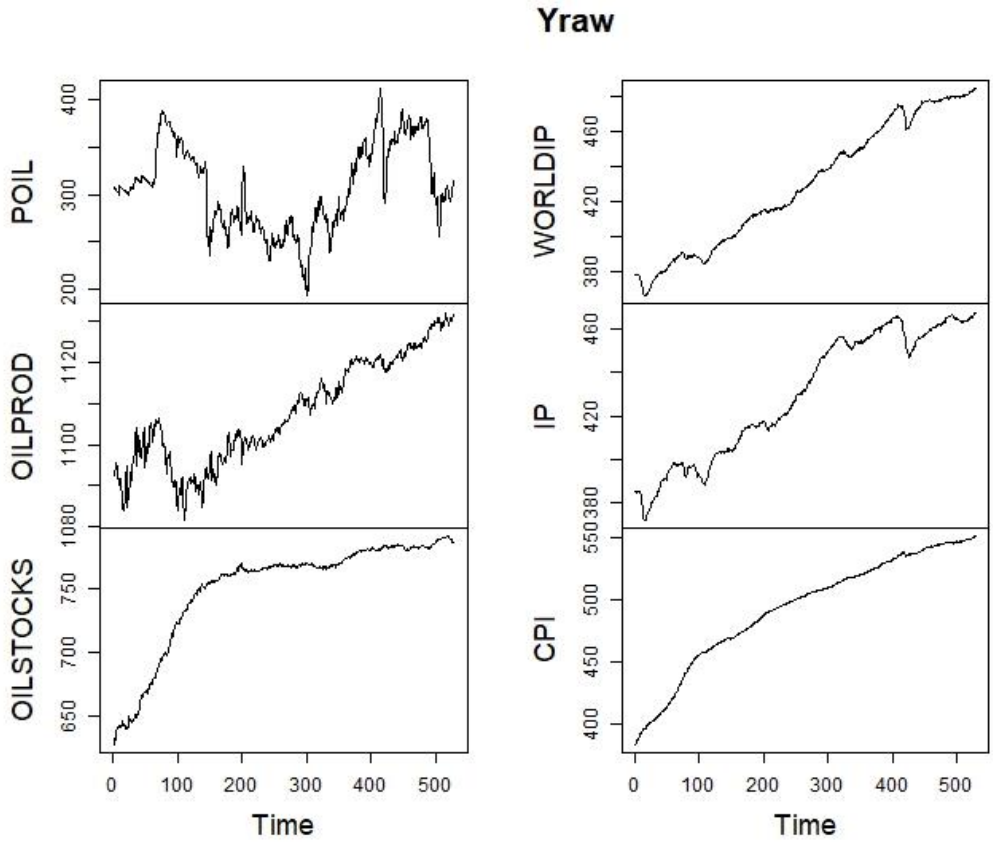
Monthly oil supply news measure constructed from 14 WTI futures contracts (matrix oilProxiesWTIM) as PC1; non-announcement months are zero. This proxy is aligned to the VAR residual sample and used only inside the identification window (1975M01–2017M12).

(4) Source

Data: Kaenzig replication files (dataBaseM.mat)

Proxy: (OilSurprisesMLog.mat)

Figure 1: Time Series of Six VAR Variables



3. Methods

We follow an external instruments VAR at monthly frequency:

(1) Reduced- form VAR (12)

Estimate an OLS VAR with a constant on the six variables over 1974M01–

2017M12; compute residuals and the companion-form multipliers.

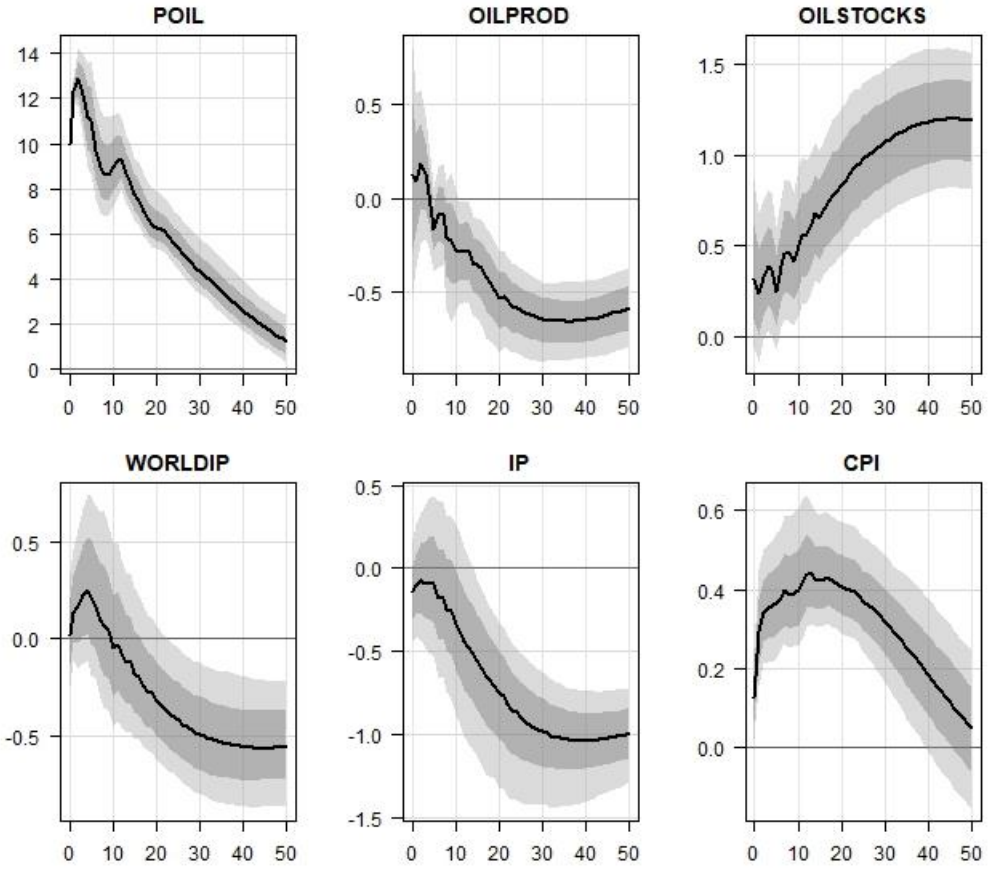
(2) External-instrument identification

The proxy is matched to the VAR residuals and restricted to the identification window. In the first stage, the oil price residual (first VAR equation) is regressed on the instrument, and the F-statistic is reported as a measure of instrument strength. In the second stage, other VAR residuals are regressed on the fitted oil price residual, and together with covariance restrictions, this delivers the impact vector that maps structural shocks to reduced-form residuals.

(3) IRFs and Uncertainty (bands)

Impulse responses are computed using the VAR's moving-average coefficients and normalized so that the impact effect on the oil price is +10. To quantify uncertainty, we implement an i.i.d. bootstrap over the identification window. For each resample, a new impact vector is estimated and IRFs are recomputed. Recentered 68% and 90% confidence bands are then constructed.

Figure 2 Impulse Responses to an Oil Supply News Shock



4. Interpretation and Results

Because our replication matches the paper's Figure 3, we follow the paper's discussion and interpretation.

Oil price (POIL). A negative oil-supply news shock produces a large, immediate rise in the real price of oil (we normalize the impact to +10%). The response decays only gradually, remaining positive for many months, consistent with agents learning about tighter future supply.

World oil production (OILPROD). Production shows little change on impact, then declines sluggishly and persistently. Together with the oil-price surge and the inventory build, this pattern is what we expect from a news shock about future supply constraints.

World oil inventories (OILSTOCKS). Inventories increase significantly and persistently. A plausible explanation is speculative/precautionary motives: if market participants interpret negative supply news as a signal of further bad news, they over-accumulate inventories to insure against future scarcity.

World industrial production (WORLDIP). Output moves little during the first year, then falls significantly and persistently. One interpretation is that oil-exporting economies benefit temporarily from higher oil receipts, masking global weakness before broader general-equilibrium effects dominate.

U.S. industrial production (IP). The decline is deeper and materializes earlier than for world IP, consistent with the U.S. historically being a large net oil importer and thus more exposed to higher oil prices.

U.S. consumer prices (CPI). CPI rises on impact and continues to increase for around one year before gradually returning toward baseline- classic cost-push pass-through from energy to prices.

The results say that both find that negative oil supply news shocks identified through OPEC-related instruments. Cause oil prices to surge, inventories to rise, production to fall sluggishly, and output and prices in the U.S., confirming that our replication matches the original study's core results and interpretations.